TWO 6H30Pi | 01 Input and controls

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

HIGH: SW1 C-H. LOW: C-L, nominal -20 dB with 10k volume loading.
Both gangs share a shaft. Turn volume down before changing gain or plugging headphones.
J1/J2 T=left, R=right, S=audio return; sleeve/chassis insulated except LK1 star bond.

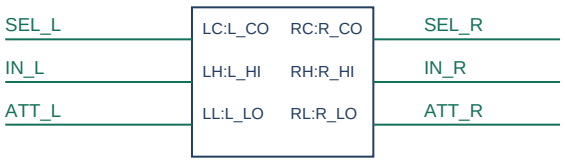
J1

3.5mm stereo INPUT [J1]

Insulated chassis jack

SW1

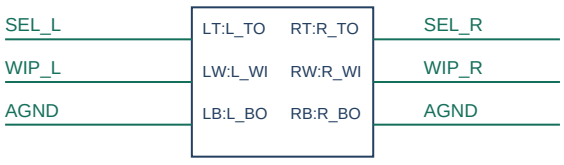
DPDT HIGH / LOW [SW1]



Break before make; two poles mechanically linked

RV1

Dual 10k log [RV1]



2 gangs; >=0.1W; gang tracking must be measured

J2

3.5mm stereo OUTPUT [J2]

Insulated chassis jack; no balanced output

R1

8.2k [RATTL]

.25W, 1%, >=250V

R2

1k [RSHL]

.25W, 1%, >=250V

R3

100k [RHPL]

.25W, 1%, >=250V

R4

8.2k [RATTR]

.25W, 1%, >=250V

R5

1k [RSHR]

.25W, 1%, >=250V

R6

100k [RHPR]

.25W, 1%, >=250V

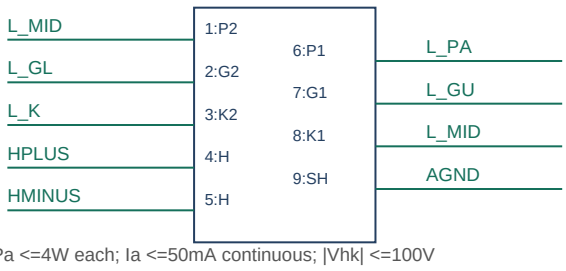
TWO 6H30Pi | 02 WCF L

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

V1 wiring view: pin 6/7/8 = upper P/G/K; 1/2/3 = lower P/G/K; 4/5 heater; 9 shield.
Nominal: P1 ~155.4V, G1 80.0V, K1/P2 82.1V, G2 0V, K2 2.49V; Iq 36.6mA.
Plate power upper ~2.68W / lower ~2.91W. CO+ faces MID. Relay does not qualify cap-short safety.

V1

6H30Pi-EB [V1]



Pa <=4W each; Ia <=50mA continuous; |Vhk| <=100V conservative

C1

4.7u [CINL]

250V, film 5%

R7

100k [RB1L]

.25W, 1%, >=250V

R8

100k [RB2L]

.25W, 1%, >=250V

C2

47u [CBL]

160V electrolytic, +/-20%

R9

47k [RGL]

.25W, 1%, >=250V

R10

100 [RGSL]

.25W, 1%, >=250V

R11

127 [RSL]

1W, 1%, metal film; nominal 0.17W

C3

4.7u [CFL]

250V, film 5%

R12

100 [RGLL]

.25W, 1%, >=250V

R13

100k [RLEAKL]

.25W, 1%, >=250V

R14

68 [RKL]

1W, 1%; nominal 0.091W

C4

2200u [CKL]

16V electrolytic +/-20%

C5

1000u [COL]

250V electrolytic +/-20%; ESR target <=0.15 ohm @1k; leakage MUST be measured

R15

100k [RDCL]

2W flameproof; defined discharge of output coupling capacitor

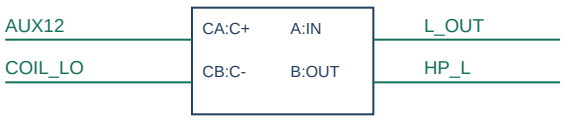
R16

10k [RBLL]

2W flameproof; 0.67W at 82V fault

K1

104-1-A-12/1 [K1]



500VDC switching, 0.5A/10W maximum; no internal diode

TP1

MID [TPLMID]

Insulated test point >=250V

TP2

K [TPLK]

Insulated test point >=250V

TP3

OUT [TPLOUT]

Insulated test point >=250V

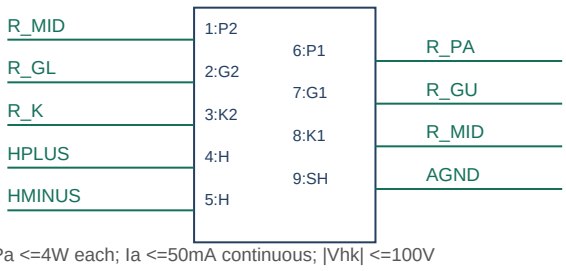
TWO 6H30Pi | 03 WCF R

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

V2 identical to V1. One complete dual triode per channel; 6.3V heaters in parallel.
WCF local feedback: plate PA -> CF -> GB -> lower grid; RK bypass CK sets AC balance.
127 ohm RS balances both model fits, not a universal tube constant. No real-headphone testing yet.

V2

6H30Pi-EB [V2]



Pa <=4W each; Ia <=50mA continuous; |Vhk| <=100V conservative

C6

4.7u [CINR]

250V, film 5%

R17

100k [RB1R]

.25W, 1%, >=250V

R18

100k [RB2R]

.25W, 1%, >=250V

C7

47u [CBR]

160V electrolytic, +/-20%

R19

47k [RGR]

.25W, 1%, >=250V

R20

100 [RGSR]

.25W, 1%, >=250V

R21

127 [RSR]

1W, 1%, metal film; nominal 0.17W

C8

4.7u [CFR]

250V, film 5%

R22

100 [RGLR]

.25W, 1%, >=250V

R23

100k [RLEAKR]

.25W, 1%, >=250V

R24

68 [RKR]

1W, 1%; nominal 0.091W

C9

2200u [CKR]

16V electrolytic +/-20%

C10

1000u [COR]

250V electrolytic +/-20%; ESR target <=0.15 ohm @1k; leakage MUST be measured

R25

100k [RDCR]

2W flameproof; defined discharge of output coupling capacitor

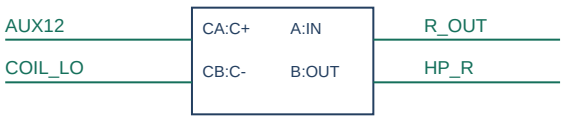
R26

10k [RBLR]

2W flameproof; 0.67W at 82V fault

K2

104-1-A-12/1 [K2]



500VDC switching, 0.5A/10W maximum; no internal diode

TP4

MID [TPRMID]

Insulated test point >=250V

TP5

K [TPRK]

Insulated test point >=250V

TP6

OUT [TPROUT]

Insulated test point >=250V

GLOBAL NET LABELS CONNECT ACROSS SHEETS | Physical pins stated; panel/relay terminals functional | NO PCB FOOTPRINTS

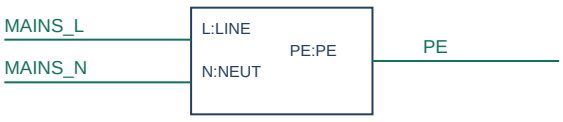
TWO 6H30Pi | 04 Mains and rectifier

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

220 VAC / 50 Hz, evaluate 198..242 VAC. Primary, HV, heater and aux are separate windings.
PE is bolted directly to chassis and transformer screen; never fused or switched.
T1 custom specification at LOAD; rated insulation, winding regulation and inrush require supplier review.

J3

IEC C14 fused inlet [J3]



250VAC certified; touch-safe enclosure

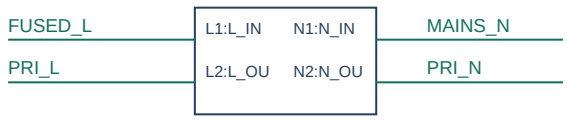
F1

T630mA H 250V [F1]

High breaking capacity; inrush validation required

SW2

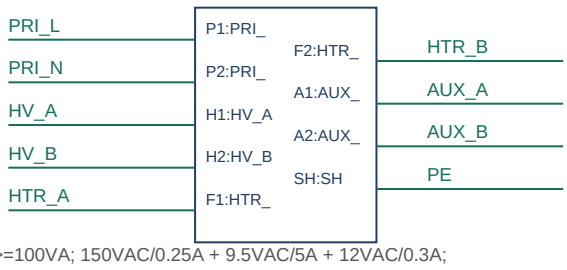
DPST mains switch [SW2]



250VAC >=2A certified

T1

220V : 150V / 9.5V / 12V [T1]



>=100VA; 150VAC/0.25A + 9.5VAC/5A + 12VAC/0.3A; reinforced isolation; thermal fuse

CH1

Dedicated PE stud [CH1]

Bolted protective-earth bond; never switched/fused

LK1

Single star bond [LK1]

Short chassis-to-signal-ground bond at single point

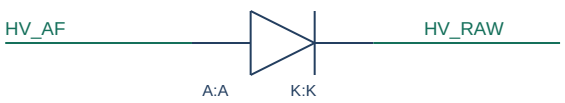
F2

T315mA H 250V [F2]

HV winding fuse

D1

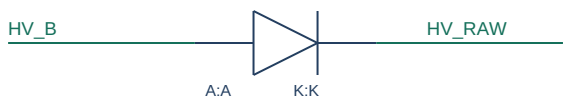
UF4007 [DH1]



1000V, 1A

D2

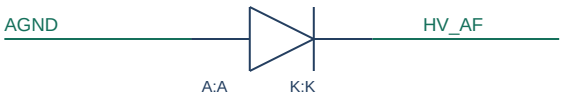
UF4007 [DH2]



1000V, 1A

D3

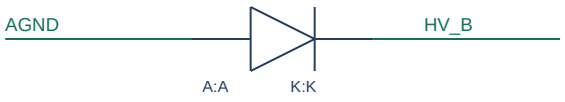
UF4007 [DH3]



1000V, 1A

D4

UF4007 [DH4]



1000V, 1A

C11

220u [CHV1]

350V +/-20%; ripple >=0.3Arms @100Hz

R27

22 [RHV]

5W pulse-rated flameproof

C12

220u [CHV2]

350V +/-20%; ripple >=0.3Arms @100Hz

R28

100k [RDIS1]

2W, >=350V, flameproof

R29

100k [RDIS2]

2W, >=350V, flameproof

TWO 6H30Pi | 05 HV regulator

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

QHV pins G/D/S = 1/2/3; tab=D. QERR/QIL pins E/B/C=1/2/3.
HV output nominal ~160V. Gate clamp DGS protects Vgs; QIL senses RIS at roughly 0.13A.
Heatsink <=8 C/W, insulated. MOSFET continuous linear SOA / loop stability are NOT qualified.

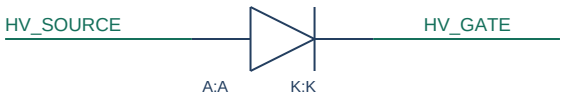
Q1
IRF820 [QHV]

500V; heatsink <=8C/W; electrically insulated tab (D)

R30
47k [RHG]

2W, >=350V

D5
12V zener [DGS]



1W

R31
4.7 [RIS]

1W; current limit ~0.13A, untrimmed

Q2
MPSA42 [QIL]

300V VCEO, 0.5A; E/B/C=1/2/3

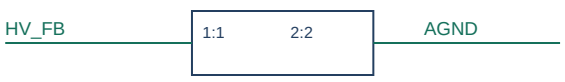
Q3
MPSA42 [QERR]

300V VCEO, 0.5A; E/B/C=1/2/3

R32
221k [RHU]

.25W, 1%, >=250V

R33
10k [RHL]

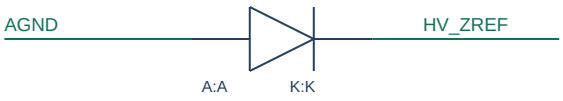


.25W, 1%, >=250V

R34
82k [RHZ]

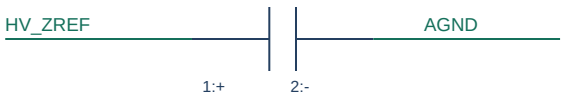
2W, >=350V

D6
6.2V zener [DZHV]



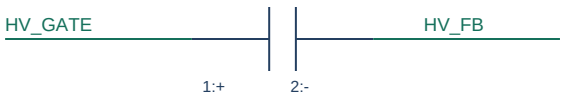
0.5W; operating current ~2mA; tolerance 2% or better

C13
47u [CZHV]



25V electrolytic

C14
10n [CCHV]



250V film; loop stability not qualified

C15
100u [CBPLUS]

250V electrolytic

R35
100k [RDIS3]

2W flameproof

TP7
B+ 160V nominal [TPB]

Insulated HV point

TWO 6H30Pi | 06 Heater and aux

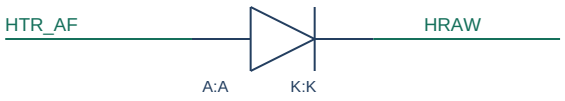
v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

Heater output is FLOATING: HMINUS ~38V above AGND. HPLUS=HMINUS+6.3V.
LM1084 pin 1 ADJ / 2 OUT / 3 IN, tab=OUT. Insulate heatsink; target <=1.5 C/W.
CHOUT aluminum ESR is intentional. Aux regulator QAUX emitter output approximately 12V.

F3
T5A 250V [F3]

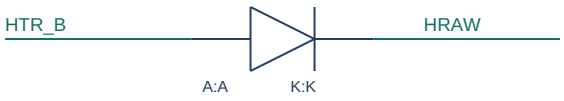
Heater winding fuse; inrush test required

D7
MBR1045 [DF1]



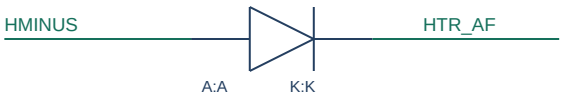
45V, 10A; insulated mounting as applicable

D8
MBR1045 [DF2]



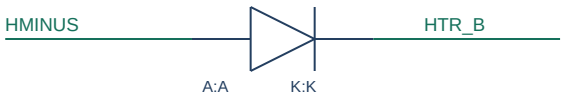
45V, 10A; insulated mounting as applicable

D9
MBR1045 [DF3]



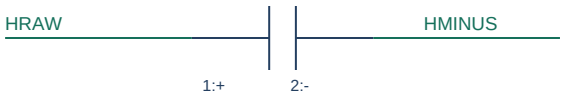
45V, 10A; insulated mounting as applicable

D10
MBR1045 [DF4]



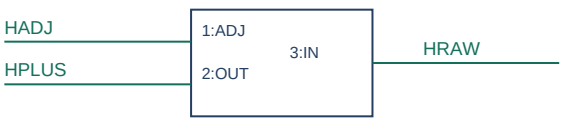
45V, 10A; insulated mounting as applicable

C16
15000u [CHF]



25V +/-20%; ripple >=4Arms @100Hz

U1
LM1084IT-ADJ [U1]



5A; heat sink <=1.5C/W; tab=OUT, insulate from chassis

R36
121 [RH1]

.25W, 1%, >=250V

R37
487 [RH2]

.25W, 1%, >=250V

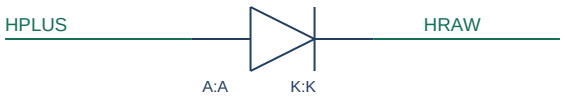
C17
220u [CHOUT]

16V aluminum, ESR 0.2..1 ohm; regulator stability verification required

C18
10u [CHADJ]

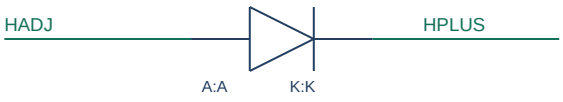
16V electrolytic

D11
1N5404 [DHREV]



400V 3A output-to-input reverse protection

D12
1N4148 [DADJ]



100V, 150mA

R38
150k [REL1]

0.5W 1%

R39
47k [REL2]

.25W, 1%, >=250V

C19
47u [CEL]

100V electrolytic

GLOBAL NET LABELS CONNECT ACROSS SHEETS | Physical pins stated; panel/relay terminals functional | NO PCB FOOTPRINTS

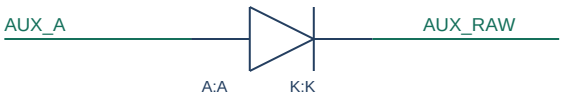
TWO 6H30Pi | 06b Auxiliary supply

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

Heater output is FLOATING: HMINUS ~38V above AGND. HPLUS=HMINUS+6.3V.
LM1084 pin 1 ADJ / 2 OUT / 3 IN, tab=OUT. Insulate heatsink; target <=1.5 C/W.
CHOUT aluminum ESR is intentional. Aux regulator QAUX emitter output approximately 12V.

D13

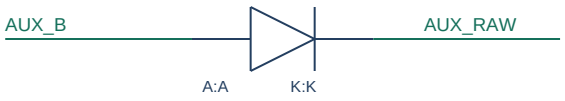
1N4007 [DA1]



1000V 1A

D14

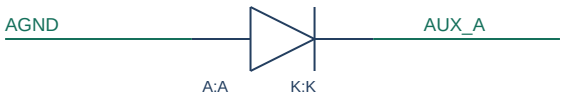
1N4007 [DA2]



1000V 1A

D15

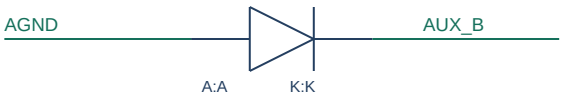
1N4007 [DA3]



1000V 1A

D16

1N4007 [DA4]



1000V 1A

C20

1000u [CAUX]

35V electrolytic

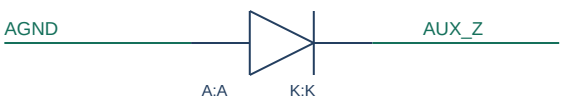
R40

470 [RAZ]

0.5W

D17

13V zener [DAZ]



0.5W 2%

Q4

MPSA42 [QAUX]

300V VCEO, 0.5A; E/B/C=1/2/3

C21

220u [CA12]

25V electrolytic

R41

4.7k [RA12]

.25W, 1%, >=250V

TWO 6H30Pi | 07 Delay and relay driver

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

Timer RT/CT with 5.1V zener and Darlington driver; typical delay must be read from simulation.
QAC/QKILL rapidly reset TIMER on missing AC pulses. Flyback uses diode + 24V zener.
DC monitors on sheets 8/9 discharge TIMER. This candidate has NOT passed capacitor-short protection.

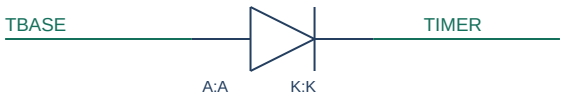
R42
330k [RT]

.25W, 1%, >=250V

C22
100u [CT]

25V low leakage; +/-20% directly changes delay

D18
5.1V zener [DZT]



0.5W

R43
1k [RTB]

.25W, 1%, >=250V

Q5
MPSA42 [QT1]

300V VCEO, 0.5A; E/B/C=1/2/3

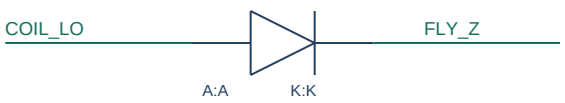
Q6
MPSA42 [QT2]

300V VCEO, 0.5A; E/B/C=1/2/3

R44
100k [RBE]

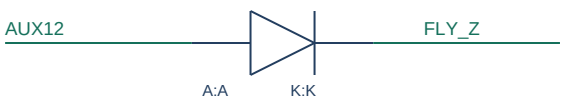
.25W, 1%, >=250V

D19
1N4148 [DFLY]



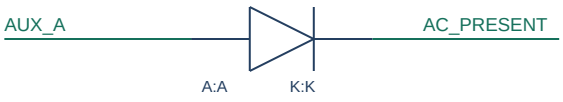
100V, 150mA

D20
24V zener [DZFLY]



1W pulse-rated; verify coil voltage and release time

D21
1N4148 [DACS]



100V, 150mA

C23
470n [CACS]

50V film

R45
10k [RACS]

.25W, 1%, >=250V

R46
47k [RACB]

.25W, 1%, >=250V

Q7
MPSA42 [QAC]

300V VCEO, 0.5A; E/B/C=1/2/3

R47
10k [RKILL]

.25W, 1%, >=250V

Q8
MPSA42 [QKILL]

300V VCEO, 0.5A; E/B/C=1/2/3

R48
2.3k [RR1]

.25W, 1%, >=250V

R49
100 [RR2]

.25W, 1%, >=250V

R50
100 [RR3]

.25W, 1%, >=250V

R51
2.3k [RR4]

.25W, 1%, >=250V

TWO 6H30Pi | 08 DC monitor L

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

DC sense = (AUX12 + L_OUT)/2; 100k/100k and 1u low-pass remove audio.
Differential pairs compare against REFHI/REFLO; reverse B-E diodes protect the transistors.
Fault path resets TIMER through diodes. Detector speed and contact stress are release-blocking tests.

R52

100k [RDS1L]

.25W, 1%, >=250V

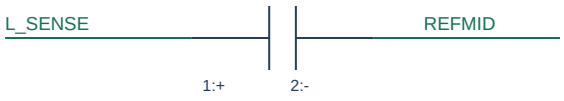
R53

100k [RDS2L]

0.5W >=250V

C24

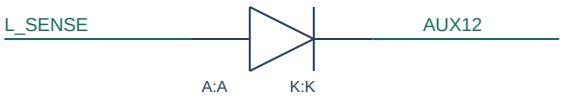
1u [CDSL]



250V film, non-polar

D22

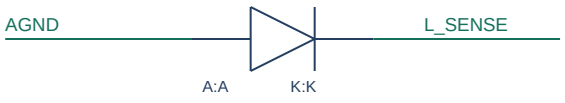
1N4148 [DSPL]



100V, 150mA

D23

1N4148 [DSNL]



100V, 150mA

Q9

MPSA42 [QPSL]

300V VCEO, 0.5A; E/B/C=1/2/3

Q10

MPSA42 [QPRL]

300V VCEO, 0.5A; E/B/C=1/2/3

Q11

MPSA42 [QNSL]

300V VCEO, 0.5A; E/B/C=1/2/3

Q12

MPSA42 [QNRL]

300V VCEO, 0.5A; E/B/C=1/2/3

R54

4.7k [RTPL]

.25W, 1%, >=250V

R55

4.7k [RTNL]

.25W, 1%, >=250V

R56

10k [RCPL]

.25W, 1%, >=250V

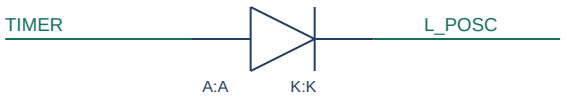
R57

10k [RCNL]

.25W, 1%, >=250V

D24

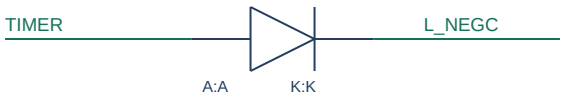
1N4148 [DTRPL]



100V, 150mA

D25

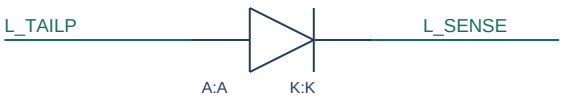
1N4148 [DTRNL]



100V, 150mA

D26

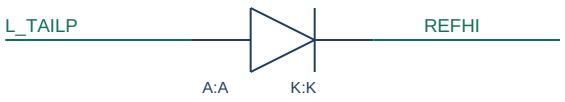
1N4148 [DBEPSL]



100V, 150mA

D27

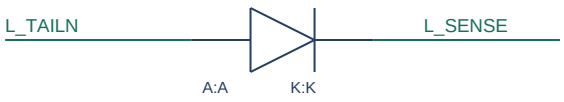
1N4148 [DBEPRL]



100V, 150mA

D28

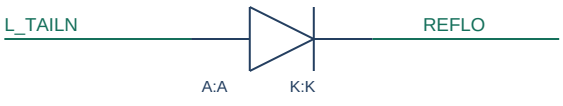
1N4148 [DBENSL]



100V, 150mA

D29

1N4148 [DBENRL]



100V, 150mA

TWO 6H30Pi | 09 DC monitor R

v0.1 CANDIDATE / NOT QUALIFIED FOR REAL HEADPHONES

DC sense = (AUX12 + R_OUT)/2. Either channel can inhibit both relay coils.
No dedicated protection IC. All detector NPNs are MPSA42, E/B/C=1/2/3.
READINESS: engineering candidate only; no PCB, no physical measurements, no real IEM connection.

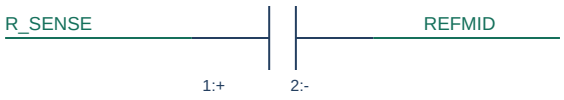
R58
100k [RDS1R]

.25W, 1%, >=250V

R59
100k [RDS2R]

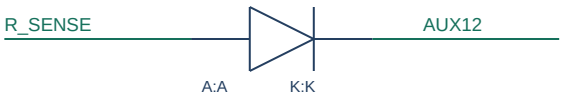
0.5W >=250V

C25
1u [CDSR]



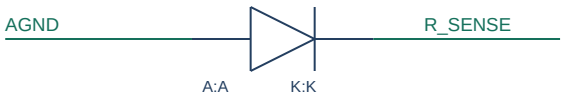
250V film, non-polar

D30
1N4148 [DSPR]



100V, 150mA

D31
1N4148 [DSNR]



100V, 150mA

Q13
MPSA42 [QPSR]

300V VCEO, 0.5A; E/B/C=1/2/3

Q14
MPSA42 [QPRR]

300V VCEO, 0.5A; E/B/C=1/2/3

Q15
MPSA42 [QNSR]

300V VCEO, 0.5A; E/B/C=1/2/3

Q16
MPSA42 [QNRR]

300V VCEO, 0.5A; E/B/C=1/2/3

R60
4.7k [RTPR]

.25W, 1%, >=250V

R61
4.7k [RTNR]

.25W, 1%, >=250V

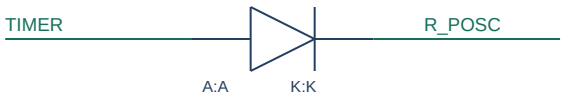
R62
10k [RCPR]

.25W, 1%, >=250V

R63
10k [RCNR]

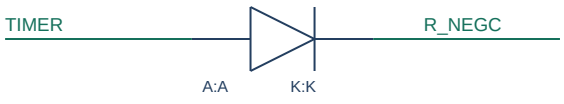
.25W, 1%, >=250V

D32
1N4148 [DTRPR]



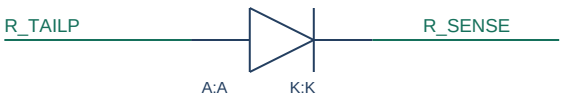
100V, 150mA

D33
1N4148 [DTRNR]



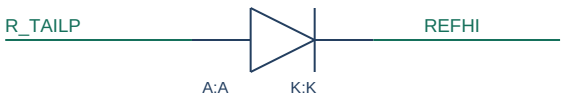
100V, 150mA

D34
1N4148 [DBEPSR]



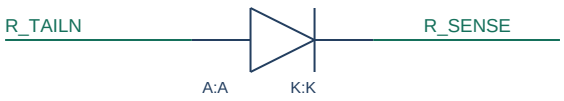
100V, 150mA

D35
1N4148 [DBEPRR]



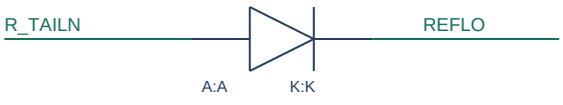
100V, 150mA

D36
1N4148 [DBENSR]



100V, 150mA

D37
1N4148 [DBENRR]



100V, 150mA